

ASMAA MOBAREK

Algiers, Algeria

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SUMMARY

AI & Data Science Engineering fourth year student at the National Higher School of Artificial Intelligence with strong interests in machine learning systems, cybersecurity, and UI/UX Design. Experienced in building machine learning systems, intelligent retrieval pipelines, and data-driven applications using Python, SQL, and modern ML tools. Passionate about solving real-world problems through scalable AI and software solutions. Eager to learn and into marketing and business management.

EDUCATION

AI & Data Science Engineering Student 2023 – Present
The National Algerian Higher School of Artificial Intelligence, Algiers, Algeria

High School Diploma - Mathematics, 18.18/20 2023
Zerrouk Bouchrit High School, Médéa, Algeria

EXPERIENCE

Elevvo Pathways - UI/UX Design Intern (Remote) Sep 2025 – Oct 2025

- Created projects including a Meditation Timer App, Pet Adoption Homepage, Weather Card, and Rating Modal
- Applied user-centered design principles to improve usability, visual hierarchy, and user experience

apeak_shop - Founder & Manager (Online Business) Jun 2024 – Jul 2025

- Managed product sourcing, listings, customer support, and order fulfillment
- Developed and maintained the brand's social media presence to drive customer engagement and sales

PROJECTS

Urban Sprawl Prediction System - Algerian Space Agency 2026

- Developed a machine learning system focused on analyzing urban sprawl patterns and predicting urban expansion trends
- Computed and analyzed urban sprawl metrics to assess spatial growth, land-use changes, and built-up area expansion on geospatial and remote sensing data.

RAG Retrieval Optimization via Quality Filtering - Qantra 2025

- Evaluated a Retrieval-Augmented Generation (RAG) pipeline using BM25 retrieval and feature engineering
- Applied clustering to identify low-quality document chunks from OCR and ground-truth datasets

SomaTrack - Student Health Analytics System 2026

- Built a machine learning pipeline to analyze the impact of study habits on student health and well-being
- Performed data collection, preprocessing, feature engineering, and predictive modeling

DNS Tunneling Detection Tool 2026

- Built a Python-based detection tool to identify DNS tunneling through traffic analysis and anomaly detection

- Analyzed DNS query patterns using entropy, frequency, and domain-level statistical features.

Suspicious Network Behaviour Detector 2026

- Developed a real-time network monitoring system to detect abnormal traffic behavior and potential threats
- Applied anomaly detection techniques to identify suspicious network activities.

NEST Hackathon - Intelligent Power & Security System 2025

- Built a real-time monitoring system for server room security and power management
- Contributed to sensor integration, system design, and rapid prototyping in a 72-hour hackathon.

Delni - Mobile Application 2025

- Contributed to mobile UI design and implementation of core application features using Flutter
- Improved usability through interface optimization and feature testing.

SKILLS

Languages: Python, SQL, C, C++, Java, Bash

ML / Data: Data Analysis, Data Mining, Data Visualization, Modeling.

Backend / Web: Flask, FastAPI, HTML, CSS, JavaScript, React

Mobile: Flutter

Databases: PostgreSQL, Oracle, NoSQL

Cybersecurity: Network Traffic Analysis, Threat Detection

Tools: Git, GitHub, Linux, VS Code, Figma

Languages Spoken: Arabic (Native), English (Fluent), French (Intermediate)

CERTIFICATES

- Figma Essentials — BYOL (2025)
- Figma Advanced — BYOL (2025)